

An Open, Reproducible Platform for Small-Area Community Indicators: The Carolinas Regional Explorer

A working paper documenting the architecture, methodology, and replication of a civic data platform for the Charlotte region

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Most community-indicator dashboards hide the statistical uncertainty of their small-area estimates and do not publish the code that produces them, so their numbers are hard to audit or reproduce. We present the Carolinas Regional Explorer (CRXP), an open platform for census-tract indicators across the 14-county Charlotte region: an open-source web application and a documented, config-driven data pipeline, joined by a versioned data contract. CRXP carries uncertainty through to the user: it propagates American Community Survey margins of error, flags low-reliability estimates, and tests change only between non-overlapping periods. It also documents choices that such platforms usually leave unstated, among them harmonizing 2010 and 2020 tract boundaries by population-weighted interpolation, detecting spatial clusters with FDR-controlled Local Moran’s I on rank-normal scores, and attributing vernacular neighborhood names to tracts from OpenStreetMap. Plain-language, cited briefs and open downloads serve non-expert users. We document the system so that other regions can replicate or adapt it with attribution.

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Note

Working paper — v0.9 (draft). This is a *living* document, updated as the platform develops; cite the version and date. The platform’s living technical documentation (application, data pipeline, methodology guide) is a companion to this paper. Citations and the reference list are managed in BibTeX (Chicago author–date). Figures are screenshots of the running application, placed beside the methods they illustrate.

1. Introduction

Communities decide where to site services, how to target investment, and whether a neighborhood’s conditions are improving on the basis of data measured at small geographic scales. A generation of “community indicator” and “neighborhood indicator” systems has grown up to meet this need, often stewarded by local *data intermediaries* that translate public data into accessible, place-based information (Sawicki and Flynn 1996; Kingsley 2014).

Yet the tools that deliver these data to the public share recurring limitations. Most present point estimates with no indication of their statistical uncertainty, even though small-area survey estimates, especially from the American Community Survey (ACS), can carry margins of error large enough to reverse apparent rankings or trends (Spielman, Folch, and Nagle 2014; Folch et al. 2016). Their methodological choices are rarely documented: how boundaries that change between censuses are reconciled, how rates are aggregated, how “change over time” is computed from overlapping survey vintages, how clusters are judged “significant.” Without that documentation, the numbers cannot be audited. And because the data-processing code is seldom released, the work is difficult to reproduce or to adapt to another region.

This paper documents the Carolinas Regional Explorer (CRXP), which addresses these limitations. CRXP is an open, reproducible platform for census-tract community indicators across the 14-county Charlotte region. It has two parts joined by a documented interface: an *open-source web application* for exploring indicators as maps, trends, and spatial analyses, and an *open data pipeline* that ingests heterogeneous public sources, computes indicators with explicit uncertainty, and hands a versioned data product to the application. We designed the platform around four commitments (*openness*, *uncertainty-awareness*, *methodological transparency*, and *replicability*). This paper serves the last two: it states the choices we made and grounds each in the relevant literature, so that others can check them.

Our contributions are:

1. **A reference implementation** (application + pipeline + versioned data contract) for open, small-area community-indicator platforms, designed so that another metro can adopt it by changing configuration (Section 7).
2. **Uncertainty carried end to end:** propagation of ACS margins of error, coefficient-of-variation reliability flags surfaced in the interface, significance-tested change between non-overlapping periods, and published county estimates with their own MOE, pooled to the region, in place of tract averages (Section 5.4, Section 5.7).
3. **Documented, literature-grounded methodology** for the hard steps: ACS 5-year comparability (Section 5.5), fractional population-weighted 2010→2020 tract harmonization (Section 5.6), direct and model-based estimation (Section 5.8), exact raster-to-tract zonal statistics (Section 5.9), and FDR-controlled spatial-cluster detection (Section 5.10).
4. **A detailed, reproducible method for attributing vernacular neighborhood names to tracts** from volunteered geographic information (Section 5.12).
5. **A config-driven, multi-source data engine** spanning survey, model-based, administrative, and satellite data, with a source-sustainability classification (Section 5.2).
6. **A replication pathway:** adoption recipe, licensing, and attribution model (Section 7).

2. Background and related work

Community indicators and data intermediaries. Planners and public administrators have compiled place-based social, economic, and environmental indicators to guide local action for decades ([Sawicki and Flynn 1996](#)). Since the 1990s, the National Neighborhood Indicators Partnership (NNIP) has organized local *data intermediaries* (university institutes, civic nonprofits, and government offices) that make neighborhood data public and build local capacity to use it ([Kingsley 2014](#)). These organizations confront a common technical problem: turning fragmented public data into trustworthy, small-area indicators that non-specialists can explore. The Charlotte-Mecklenburg Quality of Life Explorer, a long-running product of this tradition, is the direct motivation for CRXP. It is itself open source: its application and processed data are MIT-licensed, and its code has been adapted for High Point, North Carolina. CRXP extends that model from a single county’s neighborhood profile areas to census tracts across a multi-county, two-state region, and adds a documented source-to-indicator pipeline and user-facing statistical uncertainty.

Contemporary small-area platforms. Several widely used tools deliver small-area data to the public, among them the City Health Dashboard, the Opportunity Atlas, and EJScreen. The EPA removed EJScreen, its environmental-justice screening tool, from its website in February 2025; community-maintained reconstructions keep it available ([Environmental Data & Governance Initiative 2025](#)). Each tool is useful within its domain, but few combine these strengths in one platform, as Table 1 details.

2.1. How CRXP compares to existing platforms

The table contrasts CRXP with representative small-area platforms on the dimensions most relevant to transparency and replication. Entries describe each tool as publicly documented on its website and repositories as of September 2026 ([Mecklenburg County, City of Charlotte, and UNC Charlotte n.d.](#); [City Health Dashboard 2026](#); [Chetty et al. 2018](#)). The entries are descriptive, not evaluative.

Table 1: Comparison of CRXP with representative small-area data platforms.

Dimension	CRXP	Charlotte QoL Explorer	City Health Dashboard	Opportunity Atlas	EPA EJScreen (archived)
Publish grain	Census tract	Neighborhood profile area (462)	City & tract	Census tract	Block group / tract
Coverage	14-county, two-state region	Mecklenburg Co.	970+ U.S. cities	National	National
Open-source application	Yes (MIT)	Yes (MIT)	No	No	No; removed from EPA site 2025, community rebuilds

Dimension	CRXP	Charlotte QoL Explorer	City Health Dashboard	Opportunity Atlas	EPA EJScreen (archived)
Documented source-to- indicator pipeline	Yes (open code)	Processed data public; processing not published	Methods published	Methods published	Data + methods published
Margins of error shown to users	Yes (band + flag + hatch)	No	Yes (tract CIs in tables)	SEs in data; UI warnings	Percentiles, not MOE
Multi-source (survey + model + admin + satellite)	Yes	ACS + local administra- tive	Multi-source (health)	Linked tax/census records	Environmental + ACS
Time series + comparable- period change	Yes (5/10-yr, sig-tested)	Yes (trends)	Limited	Static cohorts	Limited
Interactive spatial analysis (LISA, bivariate)	Yes (FDR-LISA + correlation scatter)	Limited	Limited	No	Mapping only
Plain- language, cited indicator briefs	Yes (per indicator)	Partial	Partial	Limited	Limited
Open data download	Yes (CSV / JSON / zip)	Yes (CSV on GitHub)	Yes	Yes	Yes (archived)
Adapted or adaptable to another region	Yes (config- driven)	Yes (forked for High Point, NC)	National service	National service	National service
Proprietary GIS required	No	No	n/a	n/a	n/a

Incumbent platforms excel on coverage or in a specific domain, and the local precedent already shows that an open application can travel between regions. What none of them combines is an open application, an open source-to-indicator pipeline, and user-facing uncertainty. CRXP is built to fill that gap.

3. Design goals and principles

Six design principles put those commitments into practice. Each answers a limitation named in Section 1.

1. **Tract as the publish grain, with a regular update cadence.** The census tract is the finest scale at which most public indicators are reliably available; every CRXP indicator is published at the tract level. The catalog targets at least a biennial update cadence, and the source registry (Section 5.2) flags series that fall behind it; several do today (four NLCD land-cover measures end in 2021, LODES in 2022).
2. **Open and reproducible.** Both the application and the pipeline are open source and use no proprietary GIS, so partner organizations and students can run, audit, and extend the system. This applies the FAIR principles for the data products (Wilkinson et al. 2016) and the norms of reproducible computational research (Peng 2011).
3. **Uncertainty-first.** Because small-area estimates are uncertain, the platform carries and displays that uncertainty (Spielman, Folch, and Nagle 2014; MacEachren et al. 2012).
4. **Separation of concerns via a versioned data contract.** A data *engine* produces a documented, schema-versioned data product that an *application* consumes. That interface makes the application portable and the data reusable by other tools.
5. **Staggered updates and extensibility by configuration.** Sources update on different schedules and cover different geographies; the engine treats partial updates as normal and adds indicators through configuration.
6. **Accessible to non-experts.** Beyond the analyst-facing map, the platform offers plain-language, cited briefs and one-click data downloads, so residents and journalists can understand and reuse the data.

4. System architecture

CRXP is two components joined by a stable interface (Figure 1). The *data engine* ingests public sources, harmonizes geography, computes indicators with uncertainty, derives spatial analytics, and writes both a curated *warehouse* and the *data contract* the application reads. The *application* is a static, prerendered web client that loads the contract and renders interactive maps, trends, and analyses entirely in the browser.

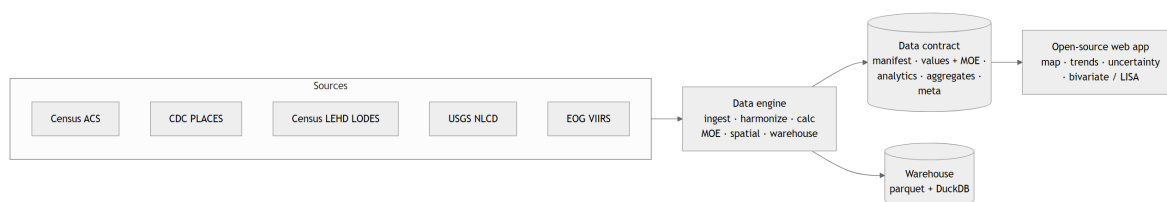


Figure 1: CRXP architecture: an open data pipeline ingests public sources and produces a versioned data contract consumed by a static web application; it also writes a reusable parquet/DuckDB warehouse.

Technology choices and rationale. The application is a static site (prerendered HTML with a vector map library) that requires no server or database at run time. Hosting is cheap, there is no

scaling work, and there is no server to secure or patch, which matters for civic projects with small budgets. The lone exception is address search, which calls a single stateless serverless function to proxy a geocoder (keeping the geocoding key server-side); the browser-side “use my location” lookup needs no server at all, resolving a coordinate to its tract by point-in-polygon against the loaded tract geometry.

Geometry carries only identifiers, and indicator values are joined to it in the browser via the map library’s *feature-state*, keeping payloads small and letting the app switch indicators and years without reloading maps. The pipeline is written in Python to draw on its data-science ecosystem; the warehouse uses columnar parquet files plus an embedded analytical database (DuckDB), giving any downstream tool, including other applications, SQL access to the same curated data without standing up a server.

5. Data pipeline and methodology

Each subsection follows one pattern: it states the choice, gives the formalism with a plain-language explanation, justifies it against the literature, and, where the choice surfaces in the interface, illustrates it with the running application. The full indicator catalog, data dictionary, and an equation reference are maintained in the companion living documentation (the Methodology Guide and Data Pipeline Technical Documentation).

5.1. Region and geography

The study region is the 14-county Charlotte metropolitan region: 11 counties in North Carolina (Anson, Cabarrus, Catawba, Cleveland, Gaston, Iredell, Lincoln, Mecklenburg, Rowan, Stanly, Union) and 3 in South Carolina (Chester, Lancaster, York), comprising 752 census tracts on 2020 boundaries. The census tract is the publish grain because it is the finest geography at which the ACS publishes the full table set and at which most administrative data can be reliably placed; tracts are designed to be relatively homogeneous units of roughly 1,200–8,000 residents ([U.S. Census Bureau 2018](#)), small enough to stand in for neighborhoods and large enough for stable estimates. County values use each source’s published county estimate, and the region is pooled from counties instead of averaged from tracts (Section 5.7). The region is a configuration value (a list of county FIPS codes) so the same engine serves any set of counties without code changes; this is what lets the platform move to another region (Section 7).

5.2. Sources and a sustainability-aware catalog

CRXP currently publishes 68 indicators drawn from five public sources and spanning eight of its ten themes (Safety and Arts & Culture remain to be populated). The U.S. Census Bureau’s American Community Survey (ACS) 5-Year Estimates supply 42 indicators: demographic, economic, educational, housing, transportation, insurance, and disability measures, plus a residential-stability measure for the Engagement theme. CDC PLACES supplies 16 model-based adult-health measures, including a recently added set of health-related social needs (food and housing insecurity, transportation barriers, social support, and loneliness). The Census Bureau’s LEHD Origin-Destination

Employment Statistics (LODES) supply 3 Economy measures from the Workplace Area Characteristics file: total jobs, job density, and a jobs–housing balance of workplace jobs per ACS household (U.S. Census Bureau 2023). The remaining seven are Environment measures from the USGS National Land Cover Database (NLCD), including Tree Canopy Cover, and EOG VIIRS Nighttime Lights (light pollution), for a total of $42 + 16 + 3 + 7 = 68$.

LODES reports jobs on 2020 census blocks, which nest exactly within 2020 tracts, so tract values are simple sums with no crosswalk. Because LODES is a synthetic, noise-infused full count rather than a sample, it carries no sampling MOE and, like the raster measures, is shown without an error band. The jobs–housing ratio’s denominator, ACS households, does carry an MOE; the platform does not yet propagate it. One ingester module per source sits behind a common interface, so adding a source (as LODES showed) or many measures from an existing source (as the PLACES expansion showed) is a localized, configuration-driven change.

Bringing your own tract data. Regions are not limited to these five sources. A generic `tract_csv` source turns any file keyed by an 11-digit census-tract GEOID into an indicator with one configuration file: the adopter names the GEOID and value columns, the data year, the kind of measure (count, proportion, rate, median, or index), and the source credit. An optional 90% MOE column passes through the same CV and reliability rules as the ACS. Files on 2010 tracts are allocated to 2020 tracts with the platform’s crosswalk: counts by allocation factor, and rates and proportions through population-weighted pseudo-counts, which require a population column. Medians and indices have no defensible allocation and must be supplied on 2020 tracts. The build stops if more than 5% of the file’s GEOIDs are not tracts of the declared vintage, since 2010 and 2020 tracts share most of their IDs and a file on the wrong vintage would otherwise leave a third of the map blank while passing the coverage check. Because the source publishes no county figure, county and regional values are derived from the tracts: population-weighted for most kinds, and summed for counts, where a county total is published only if every tract in the county has a value. The application labels these values as derived. The worked example in Section A follows one such file from download to map.

A recurring failure mode for indicator systems that aggregate many feeds is *silent staleness*: a series that quietly stops updating while still appearing current. We mitigate this with a machine-readable *source registry* that records each source’s provider, native geography, vintages, license, and update cadence, together with a *sustainability classification*: *active* (refreshed on a known cadence), *event-driven* (periodic, opportunistic), or *static / decommission-watch* (no longer updated). This classification drives the per-indicator vintage label, so users can see how current each number is.

5.3. A configuration-driven indicator model

Each indicator is declared in a small configuration file specifying its source, the variables it draws on, its *kind*, and presentation metadata. Supported kinds cover the common cases (**count**, **median**, **proportion**, **rate**, and **density**), and numerators and denominators may be sums of several variables. This matters because many community measures are *derived*: cost burden is the sum of several income-share categories; a race/ethnicity grouping sums several base counts; child poverty has a multi-line denominator. Expressing these in configuration keeps the engine small, auditable, and analyst-editable, and it makes the precise definition of every indicator inspectable without

reading source code. Variable identifiers are the official ACS codes (e.g. B19013_001E for an estimate and B19013_001M for its margin), and ACS subject tables are routed automatically to the appropriate API endpoint.

5.4. Margins of error and reliability

ACS estimates are sample-based and published with a 90% margin of error (MOE). Treating these point estimates as exact is a well-documented hazard at the tract level, where MOEs are frequently large relative to the estimate and can reverse apparent rankings or trends (Spielman, Folch, and Nagle 2014; Folch et al. 2016). CRXP therefore carries the MOE through every derived quantity and converts it to interpretable reliability information shown in the interface (Figure 2).

We first convert the MOE to a standard error and a coefficient of variation:

$$SE = \frac{MOE}{1.645}, \quad CV = \frac{SE}{|\hat{x}|} \times 100\%,$$

where $\hat{x}$ is the estimate (the 1.645 factor is the 90% normal quantile the Census uses). *In plain terms*, the standard error is the typical sampling “wobble” in the estimate, and the CV expresses that wobble as a percentage of the estimate, a unit-free measure of reliability. We flag estimates as *ok* ($CV \leq 15\%$), *caution* ($15\% < CV \leq 30\%$), or *unreliable* ($CV > 30\%$), following long-standing Census practice and the National Research Council’s guidance on using the ACS (Citro and Kalton 2007; U.S. Census Bureau 2020). A known limitation is that the CV is unstable for near-zero estimates, where a tiny absolute MOE can produce a huge CV; we report such values unsuppressed.

For derived measures we propagate the MOE using the Census Bureau’s published formulas. For an *aggregate* (a sum of estimates), MOEs combine in quadrature:

$$MOE_{\text{sum}} = \sqrt{\sum_i MOE_i^2}.$$

For a *proportion* $p = \hat{n}/\hat{d}$ whose numerator is a subset of the denominator,

$$MOE_p = \frac{1}{\hat{d}} \sqrt{MOE_n^2 - p^2 MOE_d^2},$$

falling back to the ratio formula below when the radicand is negative (the documented Census behavior). For a *ratio* $R = \hat{n}/\hat{d}$ whose numerator is *not* a subset of the denominator,

$$MOE_R = \frac{1}{\hat{d}} \sqrt{MOE_n^2 + R^2 MOE_d^2}.$$

In plain terms, when we combine estimates the uncertainties add up, but not linearly: independent errors partly cancel, which is why they combine through squares and square roots. These propagation rules are implemented as pure functions and unit-tested against the worked examples in the Census handbook.

Finally, to compare two estimates we use the Census difference test: the difference is statistically significant at 90% confidence when

$$\frac{|\hat{x}_1 - \hat{x}_2|}{\sqrt{SE_1^2 + SE_2^2}} > 1.645.$$

This guards against reading noise as change (Section 5.5). The interface puts all of this at the point of use: a selected tract’s trend shows a shaded $\pm$ MOE band, a reliability badge, and the significance-flagged change (Figure 2).

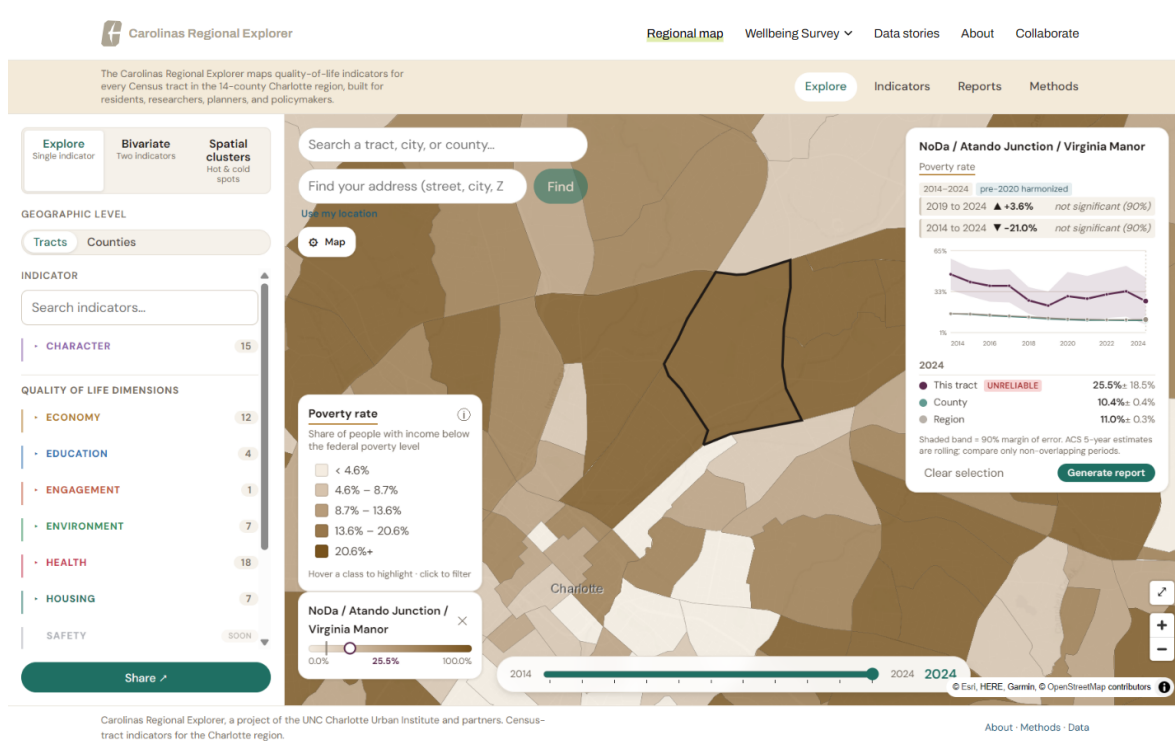


Figure 2: Uncertainty at the point of use: a selected tract’s trend with a $\pm 90\%$ margin-of-error band, an ok/caution/unreliable reliability badge, the significance-tested 5- and 10-year change, and provenance badges (vintage; *model-based* for CDC PLACES; *pre-2020 harmonized* for ACS).

CDC PLACES estimates are model-based and published with a 95% confidence interval rather than a 90% MOE; we convert the CI half-width to a 90%-equivalent MOE so reliability is comparable across sources, while flagging such indicators *model-based* in the interface because a model-based interval and a sampling MOE are different kinds of uncertainty.

5.4.1. Temporal comparability of model-based estimates

Uncertainty also limits which trends a platform may show, and here the sources diverge sharply. The PLACES health measures are assembled into a multi-year series by stitching six releases (newest-release-wins, with pre-2024 rows harmonized onto 2020 tracts), but that series is presented strictly as levels per year, never as change. CDC is explicit that “the current modeling procedure does not support using the estimates to track changes at the local level over time” (Centers for Disease Control and Prevention 2024b). Sub-county estimates poststratify to a fixed population structure, and time is not a model variable even at county level. Apparent movement is often an artifact of questionnaire changes (CDC flags colorectal-screening prevalence specifically) or of pandemic-era 2020 BRFSS collection. The platform therefore treats temporal comparability as a source-level

property carried in the data contract: a `crossReleaseTrend` flag marks every PLACES indicator, and each interface that would otherwise assert change (the significance-tested change markers, the printable report’s *change* column, and the county-profile deltas) is suppressed for those indicators, while the level series and its uncertainty band remain. This is the same uncertainty-first principle applied one level up: a platform that propagates margins of error should also refuse to manufacture a trend its source cannot support. For survey-based ACS indicators, by contrast, change *is* meaningful when read correctly, which the next section addresses.

5.5. ACS comparability over time

ACS 5-year estimates *overlap*: the 2020–2024 and 2019–2023 estimates share four years of sample, so adjacent vintages are not statistically independent and their difference is not a valid measure of change (Citro and Kalton 2007; U.S. Census Bureau 2020). CRXP therefore treats the annual series as a rolling 5-year estimate and computes change and significance only between non-overlapping periods. For a selected tract the application reports two changes: a 5-year (latest minus five) and a 10-year (latest minus ten), each evaluated with the difference test in Section 5.4. These endpoints are derived from the latest available year, so they advance automatically as new data is released (adding a 2025 vintage shifts the comparisons to 2020→2025 and 2015→2025 with no code change). The trend chart’s x-axis is drawn proportional to the actual year, so uneven cadence (biennial NLCD releases, periodic VIIRS) and gaps appear at their true spacing.

5.6. Harmonizing 2010 and 2020 tract boundaries

Census tract boundaries are redrawn each decade; ACS vintages through 2019 use 2010 tracts, and 2020 onward use 2020 tracts. In this region, the 611 tracts of 2010 became 752 tracts in 2020, and 135 of the 2010 tracts were split or redrawn (fewer than 99% of their residents fall in a single 2020 tract). Comparing a 2017 value to a 2024 value therefore means comparing two different geographies. To place the series on a single 2020 geography we *harmonize* the earlier years by areal interpolation (Goodchild and Lam 1980), weighting by population rather than land area.

Using the Census 2010 2020 *block* relationship files, each 2020 block’s population is apportioned across the 2010 tracts it overlaps in proportion to the overlap land area (a dasymetric refinement that splits a straddling block rather than assigning it whole to one side); those apportioned block populations are then summed to form the allocation factor from a 2010 tract to a 2020 tract:

$$afact(t_{10} \rightarrow t_{20}) = \frac{\text{2020 block population apportioned to } (t_{10} \cap t_{20})}{\text{2020 block population apportioned to } t_{10}}.$$

Allocation is applied to the *raw variables before indicators are computed*: counts are summed, $count_{20} = \sum_{t_{10}} count_{10} \cdot afact$; MOEs combine as $MOE_{20} = \sqrt{\sum (afact \cdot MOE_{10})^2}$ (an independence approximation); medians, averages, and indices are population-weighted averages of the contributing tracts’ values.

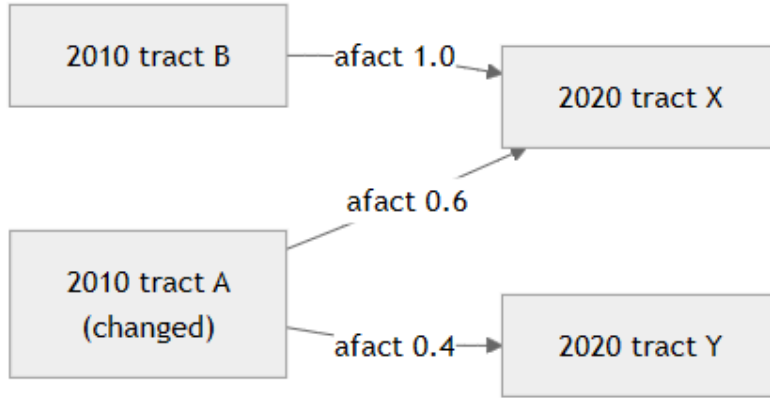


Figure 3: Population-weighted allocation: a changed 2010 tract’s value is split across the 2020 tracts that absorb its population (allocation factors sum to 1 per 2010 tract).

Why population, not area. Area weighting assumes people are spread uniformly within a tract, which badly misallocates tracts that split into pieces of widely different density. In this region, a small-but-dense 2020 tract that holds roughly a fifth of an old tract’s population receives only ~0.4% of it under area weighting but its true ~21% under population weighting.¹ Population-based interpolation is the standard for count-type variables and underlies established tract-harmonization products (Logan, Xu, and Stults 2014; Manson et al. 2023); dasymetric refinements with ancillary data extend it (Mennis 2003; Eicher and Brewer 2001). This harmonization raised the coverage of valid comparable-period change from roughly 480 to about 749 of 752 tracts. Medians remain approximate: they are population-weighted averages of tract medians, which is exact only when within-tract distributions are similar; the unbiased fix (reconstructing medians from afact-weighted binned distributions) is planned.

5.7. County and regional estimates

A county or regional value is a different estimand from a tract value, and averaging tract values to obtain it is biased; for medians and indices it is statistically invalid. The unweighted mean of tract rates over-weights small tracts [the *ecological* and *modifiable areal unit* problems; Robinson (1950); Openshaw (1984)]; a county’s median income or Gini coefficient cannot be recovered by averaging its tracts’ medians at all. CRXP therefore does not aggregate tracts for county metrics. Instead it uses the county-level estimate published by each source, and pools the region explicitly.

County. Because county boundaries are stable across the 2010 and 2020 censuses (no harmonization needed), the engine fetches each county’s published estimate directly (ACS county tables carrying their own, smaller county margin of error; the CDC PLACES county model; or county zonal statistics for the satellite measures) and runs them through the *same* indicator calculator used for tracts. Each county thus gets its published value and MOE for every kind, including the medians and indices that averaging tracts would get wrong.

¹2010 tract 37007920300 in Anson County: its 2020 successor tract 37007980000 covers 0.45% of the old tract’s land area but holds 20.9% of its population.

Region. The 14-county region has no single published estimate, so it is pooled by indicator kind: counts are summed; rates use the pooled numerator over the pooled denominator,

$$p_{\text{region}} = \frac{\sum_c \text{num}_c}{\sum_c \text{den}_c},$$

(the exact pooled rate, with its Census proportion/ratio MOE on the summed components); and medians and indices use a population-weighted mean of the county values. That is an approximation: an exact regional median would require microdata. County and regional series, with propagated MOE (`countyMoe/regionMoe`), are precomputed into a `county_indicators` table and exported in the contract, so the interface labels these lines plainly as *County* and *Region*. (The one place a simple unweighted mean remains is the *ad-hoc average of a user-selected set of tracts* in a custom report, which is a selection summary, not a published geography.)

5.8. Direct and model-based estimates

Most CRXP indicators are *direct* estimates published by the ACS at the tract level, the simplest and most defensible kind. Where a measure is modeled rather than tabulated, we use *model-based small-area estimates*: the CDC PLACES health measures are produced by multilevel regression and poststratification (MRP) on survey data, a now-standard approach for small-area prevalence (Gelman and Little 1997; Centers for Disease Control and Prevention 2024a). For cross-tabulations the ACS does not publish at tract level, a planned tier will fit small-area models to public-use microdata with design-based standard errors in the Fay–Herriot tradition (Fay and Herriot 1979; Rao and Molina 2015). A companion review compares the methods available for that tier, from allocation-factor crosswalks to model-based estimation, when the microdata geography (the PUMA) does not nest into counties (Venkitasubramanian 2026c). Every indicator records its estimation method in the warehouse and metadata, so users and reviewers can see *how* each number was produced and treat direct, model-based, and (eventually) survey-modeled estimates with appropriate care.

5.9. Raster-to-tract zonal statistics

Environment indicators are derived from 30-meter satellite rasters via area-weighted zonal statistics computed with the *exact fraction* of each pixel falling inside a tract, rather than the cruder pixel-centroid rule. This is the `exactextract` method (Baston 2025). The centroid rule misassigns partial pixels along tract edges, and irregular, mid-sized polygons like tracts have many edge pixels. Land-cover *shares* (forest, cropland, wetlands, developed land) are the coverage-weighted fraction of a tract’s non-nodata area in the relevant NLCD classes; *continuous* measures (impervious surface, tree-canopy cover, nighttime-light radiance) are coverage-weighted means (NLCD: Homer et al. 2020; VIIRS: Elvidge et al. 2021). The Annual NLCD product supplies land cover for most years from 2001 onward, so these indicators carry a multi-year series, not only decadal snapshots.

Because these are wall-to-wall measurements, not samples, they carry *classification accuracy* instead of a sampling MOE, and the application shows no error band or reliability flag for them, because the uncertainty is of a different kind. Two caveats apply and are documented: satellite land-cover products have published per-class accuracy that varies by class, and any tract-level zonal statistic inherits the arbitrariness of the tract boundary itself [the modifiable areal unit problem; Openshaw (1984)].

5.10. Spatial analytics: standardization, classification, and clusters

For each indicator and year the engine computes three things consumed by the interface.

Standardization. Per indicator-year, values are converted to z-scores (mean 0, sd 1 across tracts), which the bivariate scatter and cluster analysis build on.

Classification. The choropleth uses quantile class breaks computed once over the pooled multi-year distribution and held constant across years, so a given color means the same value range in every year and change over time can be read off the map. Choosing and fixing a classification is a consequential cartographic decision (Slocum et al. 2009; Brewer 2016); fixing it across time is what keeps the animated year slider from misleading.

Clusters and outliers. Neighborhood data are spatially structured: “everything is related to everything else, but near things are more related” (Tobler 1970). We detect that structure with Local Indicators of Spatial Association [Local Moran’s I; Anselin (1995)], which classify each tract as part of a High–High or Low–Low *cluster*, a High–Low or Low–High *spatial outlier*, or *not significant* (Figure 4). Significance is a conditional-permutation pseudo p-value: tract i ’s value is held fixed while its hypothetical neighbors are drawn *without replacement* from the *other* tracts’ scores (9,999 permutations). This is the correct reference distribution for the conditional test, and a correction over a naïve with-replacement draw. Because 752 tracts are tested simultaneously, the pseudo p-values are then Benjamini–Hochberg false-discovery-rate controlled (Benjamini and Hochberg 1995) before a tract is called significant; without this correction, dozens of tracts would light up by chance alone.

Two further choices make the corrected test usable in practice. First, the test runs on *rank-based inverse-normal scores*, $\Phi^{-1}(r_i/(n+1))$ for a tract of rank r_i among n , rather than on raw values (Beasley, Erickson, and Allison 2009). Many community indicators are strongly right-skewed: a few tracts sit seven to nine standard deviations above the regional mean, and because every permutation can draw them, they widen the null distribution until ordinary clusters cannot beat it. For poverty rate in 2024, which has a clear global spatial autocorrelation (Moran’s $I \approx 0.31$), 229 tracts have an unadjusted $p < 0.05$ but only 7 remain significant after FDR correction on raw values; on rank-normal scores, 153 do. *In plain terms*, the test asks whether a tract and its neighbors rank unusually high (or low) *together* in the region, so a handful of extreme tracts cannot mask real clusters; High and Low therefore refer to rank position.

Second, the permutation count sets the smallest attainable p-value, $1/(P+1)$. The BH procedure over m tests rejects nothing until some p-value falls below α/m , which is about 6.6×10^{-5} here; with $P = 999$ (floor 0.001) at least 15 tracts had to share the floor before any was called significant, and several strongly clustered indicators showed no clusters at all. With $P = 9,999$ the floor is 10^{-4} , and two tracts at the floor suffice. The rank transform has a cost: a High–High cluster says that a group of tracts ranks high together, not that their values are far above the regional mean, so in an indicator with a narrow range a significant cluster can be substantively small. Results are also not comparable with a raw-value LISA run in standard packages.

Spatial weights are k-nearest-neighbor ($k = 8$), row-standardized, built from each tract’s *representative point* (a point guaranteed to lie inside the polygon). A bounding-box midpoint, by contrast, can fall outside an L-shaped or coastal tract and distort the neighbor graph. LISA and z-scores are computed once in the pipeline (the single statistical source of truth) and consumed by the app

as precomputed values. Clusters are computed for census tracts only; the county view offered for the choropleth is not available in cluster mode.

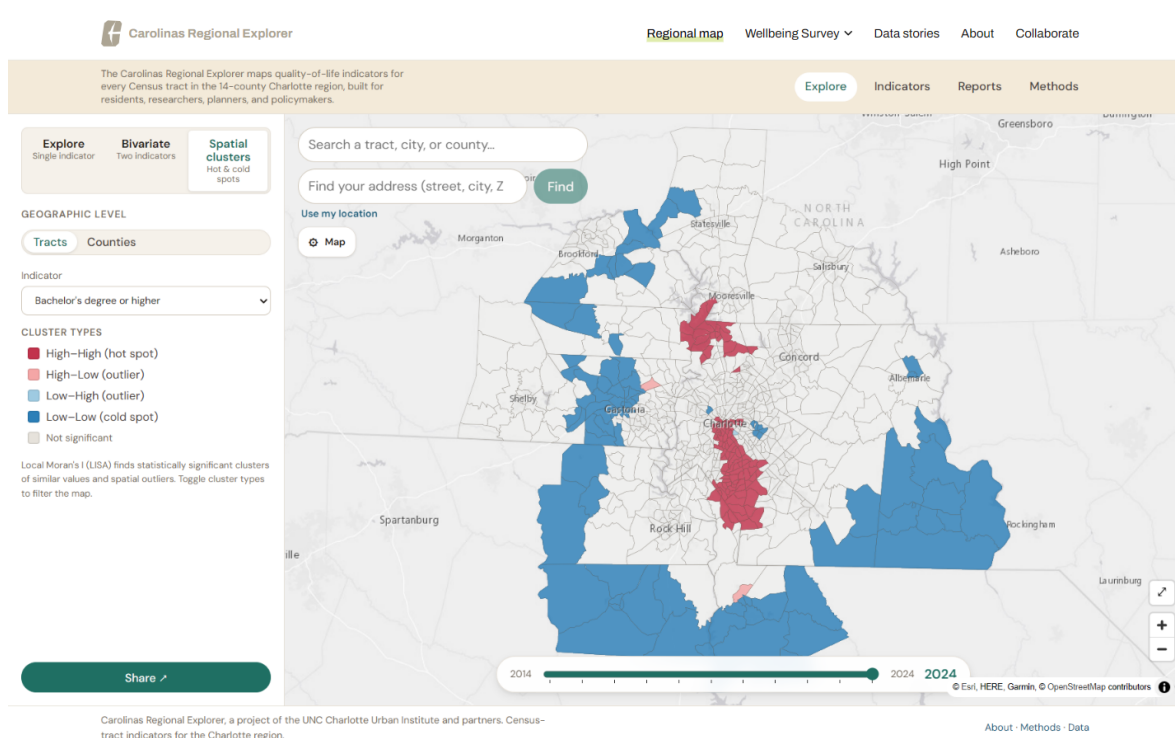


Figure 4: Spatial clusters (LISA) for bachelor’s-degree attainment, 2024: High–High and Low–Low clusters and High–Low / Low–High spatial outliers, after FDR correction.

5.11. Bivariate analysis

To examine how two conditions co-occur, such as poverty and educational attainment, the application crosses two indicators on a 3×3 bivariate choropleth: each indicator is split into rank terciles and the nine combinations are given a two-dimensional color scheme, a technique with a long cartographic pedigree (Olson 1981; Brewer 2016). Hovering a grid cell highlights the matching tracts. Like the cluster map, the bivariate view is available for census tracts only.

Beside the map, a correlation scatter plots the two indicators’ standardized values, one dot per tract colored by its bivariate class, with marginal distributions, a least-squares regression line, and both Pearson and Spearman correlations (Figure 5). Spearman (rank) is shown alongside Pearson because most indicators are skewed and the choropleth already colors by rank. Both are presented as descriptive only: tract observations are spatially autocorrelated, so the *effective* sample size is far below the tract count and a naïve significance test would be badly anti-conservative (Clifford, Richardson, and Hémon 1989). The scatter therefore reports no p-value, and two-way hover links each dot to its tract on the map and names the neighborhood (Section 5.12).

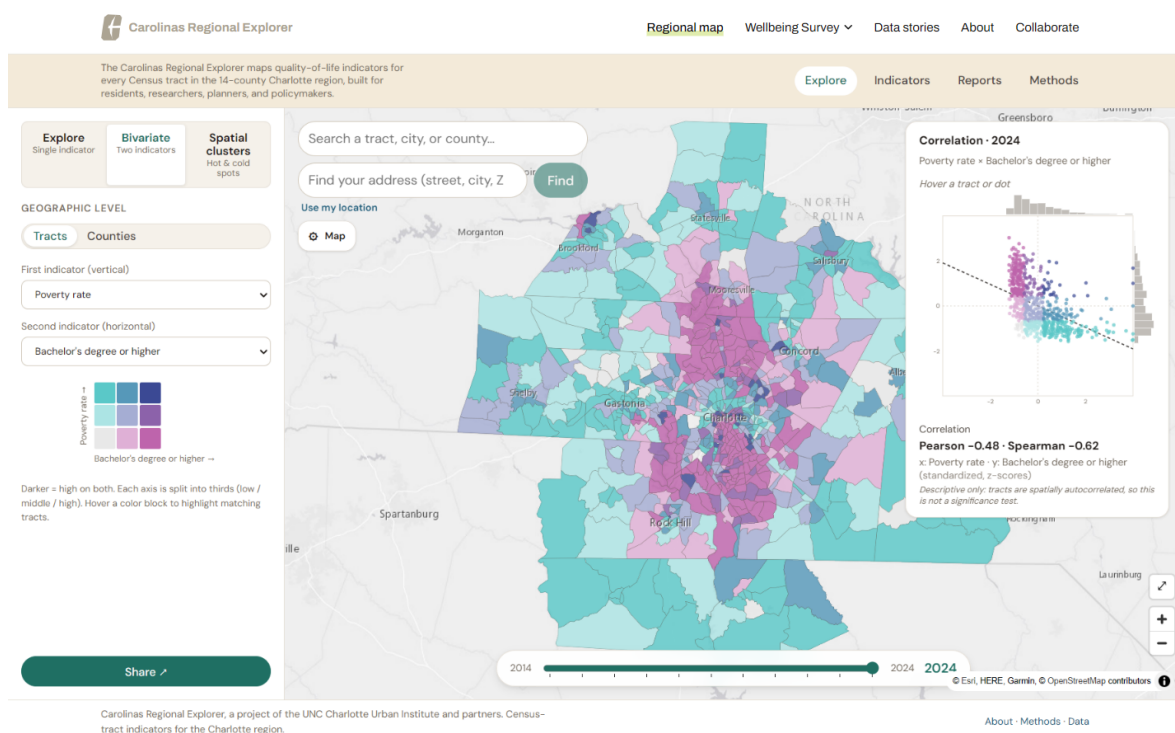


Figure 5: Bivariate analysis: a 3×3 tercile choropleth (poverty rate × bachelor’s-degree attainment) with a companion correlation scatter of standardized values — marginal distributions, regression line, and descriptive Pearson and Spearman correlations.

5.12. Attributing vernacular neighborhood names to tracts

Residents think in *neighborhoods*, not tract numbers, but neighborhoods are *vernacular regions* with no official boundaries; their extents are fuzzy, overlapping, and contested (Montello et al. 2003). To help users orient themselves, CRXP attaches one or more approximate neighborhood names to each tract, labeled as for orientation only. The method has four steps, each reproducible.

- 1. Acquire named points (volunteered geographic information).** We query the OpenStreetMap (OSM) Overpass API for `place = neighbourhood | suburb | quarter` point features within the region’s bounding box. OSM is a mature source of volunteered geographic information whose place data are broadly fit for orientation, with the usual caveat of uneven contributor coverage (Goodchild 2007; Haklay 2010). Results are cached so builds are deterministic and offline-repeatable.
- 2. Region-filter and de-duplicate.** Each point is tested against the 14 county polygons (point-in-polygon); points outside the region are dropped, and duplicates are removed by (`name`, `county`) so the same name in two counties is kept distinct. This yields the committed `neighborhoods.json`.
- 3. Attribute tracts by Voronoi area-weighting.** Each tract is assigned a *share* of each nearby neighborhood by approximating its overlap with the Voronoi (Thiessen) tessellation of the neighborhood points (Okabe et al. 2000), the partition of space into “the area closest to each named point.” Concretely, we lay a deterministic grid over the tract’s bounding box, keep the interior sample points (point-in-polygon), and assign each sample to its nearest neighborhood point; the

fraction of a tract’s interior samples nearest to a given neighborhood is that neighborhood’s *area share* of the tract. The result is a ranked list of neighborhoods with shares for each tract, since a tract often spans more than one named area.

4. Threshold, rank, and fall back. We keep neighborhoods whose share is at least 15%, up to three per tract, ranked by share. To avoid attaching a far-away name to a tract in a sparsely-mapped rural area, we compute the great-circle (haversine) distance from the tract’s centroid to its nearest neighborhood point and apply an 8 km distance cap: beyond it, or when no neighborhood clears the share threshold, the tract falls back to its city/place name (point-in-polygon against municipal boundaries) and then its county. The display label is “*Name* area” for a single neighborhood or “*A* / *B*” for several. About four in five tracts (599 of 752) receive a neighborhood label; the rest fall back to a city or town (30) or their county (123). The parameters (grid resolution, 15% share floor, three-name cap, 8 km cap) are configuration values, and the whole step is pure and unit-tested.

A region that keeps its own neighborhood points (a planning department’s list, for example) can use them instead of OSM: the same attribution runs on any GeoJSON file of named points (`gen-neighborhoods.mjs --points <file> --name-field <field> --attribution "<credit>"`), and the credit replaces the OSM notice on the Methods page and in the download bundle.

These names are approximate and for orientation only, and the interface says so; they are not official boundaries, and OSM coverage varies. The attribution is dasymetric in spirit ([Mennis 2003](#)): it distributes a fuzzy, point-based signal over areal units using an explicit distance-based rule, without asserting precise borders.

5.13. Warehouse and the versioned data contract

Computed indicators are stored in a long-format `tract_indicators` table (geoid, indicator, year, value, MOE, CV, reliability, period window, rolling flag, estimation method, source vintage) as one parquet file per indicator plus a DuckDB database that any tool can query with SQL. The parquet files are committed to the repository so the contract can be rebuilt on a fresh clone without re-fetching from upstream APIs.

On *handoff*, the engine emits the application data contract and merges it into the app: an indicator manifest (carrying a `schemaVersion`), per-indicator value files (with MOE/CV/reliability), pre-computed spatial analytics (z and FDR-LISA), county/region aggregates (with propagated MOE), per-indicator metadata, and a `provenance.json` recording the library and GDAL/PROJ versions that produced the data. Manifest and aggregate updates are written atomically (temp file + rename) so an interrupted handoff cannot leave a half-written contract. The contract is the only coupling between engine and application, and its `schemaVersion` lets the app refuse data it does not understand.

5.14. Quality assurance and testing

Quality control is layered. Ingestion is defensive: ACS requests retry with backoff, and a *partial* failure (some counties returned, others did not) raises an error instead of silently publishing a truncated year that would corrupt the series. Only a vintage that is absent for every county is skipped. A build-time QA gate fails the build if a year’s geographic coverage collapses, its

null rate is excessive, or finite values fall outside a configured plausible range. Each check also leaves a record: a JSON report per indicator and a summary table in `warehouse/qa/` (Figure 6), written for failed checks as well as passing ones. `python -m crxp_data.qa --all` re-runs the gate over the committed warehouse without fetching anything, and the reports ship with the public repository. The statistical core (MOE propagation, the difference test, the harmonization allocation, representative-point centroids, and the FDR procedure, verified to control false discoveries on spatially random data) is unit-tested against worked examples. Finally, the application’s contract validator checks the handed-off files before every build: supported `schemaVersion`, value-years aligned to the manifest, numeric sanity (no NaN), analytics content (LISA quadrants in range, years aligned), aggregate completeness, and geoid reconciliation.

CRXP QA report

Generated 2026-10-05. Each indicator's tract table is checked per year for tract coverage, the share of missing values, and (where configured) a plausible value range.

ID	Indicator	Source	Years checked	Min coverage	Max null rate	Range	Status
1	Older adults (65+)	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
2	Youth (under 18)	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
3	Veterans	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
4	Asian residents	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
5	Black residents	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
6	Hispanic or Latino residents	acs	2014–2024 (11)	99.9%	0.4%	—	PASS
40	No health insurance (adults 18–64)	cdc_places	2018–2023 (6)	99.6%	0.0%	—	PASS
41	Obesity (adults)	cdc_places	2018–2023 (6)	99.6%	0.0%	—	PASS
42	Diabetes (adults)	cdc_places	2018–2023 (6)	99.6%	0.0%	—	PASS
43	Frequent mental distress (adults)	cdc_places	2018–2023 (6)	99.6%	0.0%	—	PASS
70	Forest cover	raster	2001–2021 (9)	100.0%	0.0%	—	PASS
71	Cropland & farmland	raster	2001–2021 (9)	100.0%	0.0%	—	PASS
72	Wetlands	raster	2001–2021 (9)	100.0%	0.0%	—	PASS
75	Light pollution (night lights)	viirs	2014–2024 (3)	100.0%	0.0%	—	PASS
82	Total jobs (workplace)	lodes	2014–2022 (9)	99.9%	0.0%	—	PASS
83	Job density	lodes	2014–2022 (9)	99.9%	0.0%	—	PASS
84	Jobs–housing balance	lodes	2014–2022 (9)	99.9%	36.0%	—	PASS

... 51 more indicators not shown; all PASS

68 indicators checked; 0 failed.

Figure 6: The QA summary for all 68 indicators, regenerated from the committed warehouse: years checked, the lowest tract coverage and highest missing-value rate across those years, any configured value range, and pass or fail.

6. The application

The application turns the data contract into an explorable interface for several audiences at once: an analyst’s map, a resident’s plain-language explainer, and an open-data portal. The public site embeds the explorer inside a wider regional site (the map figures show that site’s navigation), while profile, report, and data pages are served directly by the application.

Explorer. The headline view is a choropleth explorer (Figure 7): an indicator mapped at tract or county level with the cross-year-consistent class breaks of Section 5.10, an interactive legend that filters the map by class, a year slider, and the uncertainty and comparable-period machinery of Section 5.4 surfaced on selection. Indicators are organized by theme, alphabetical within each theme, and a sub-heading separates *Character* (general demographics) from the *Quality-of-Life dimensions*; indicator dropdowns are grouped into theme-colored sections so a long list is easy to scan.

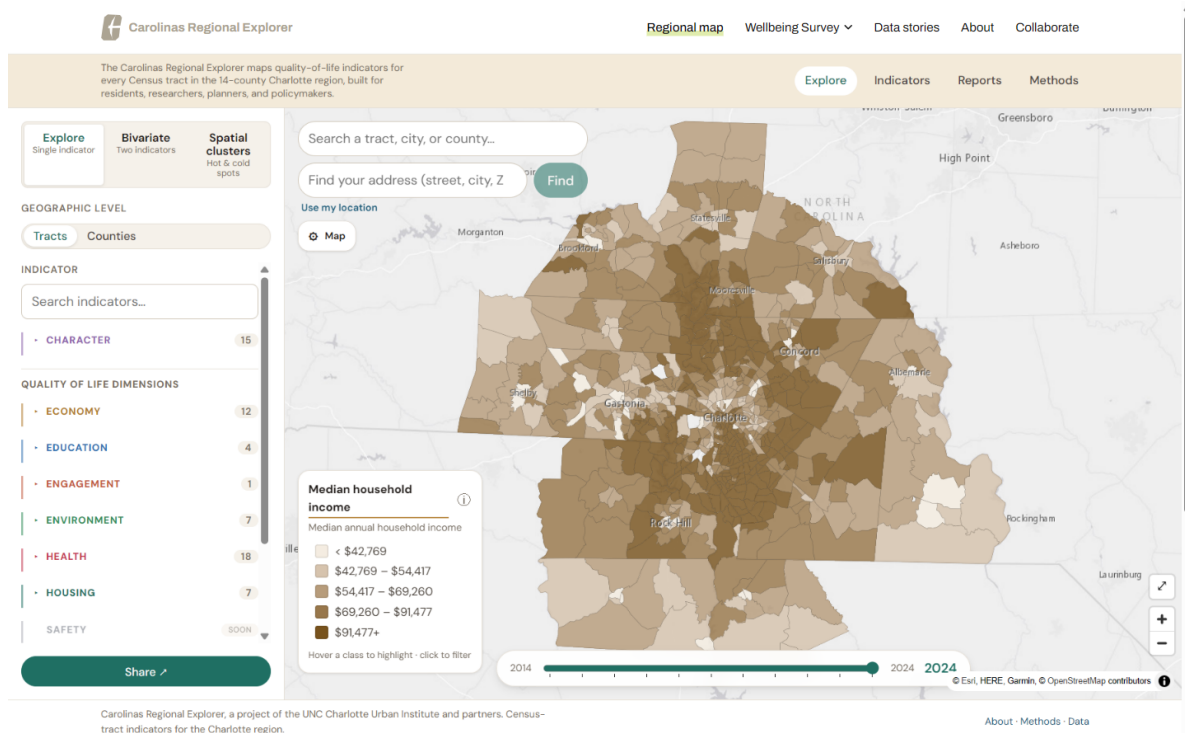


Figure 7: The choropleth explorer: median household income by census tract, with the theme-colored class legend, geography toggle, analysis-mode bar, and year slider. Bivariate and spatial-cluster modes share this canvas.

Plain-language, cited briefs. Every indicator has a “*why this matters*” brief written for residents and journalists, with a short list of verified “Learn more” resources and an “About the data” provenance note. The explore sidebar shows a brief lead with a link to the full indicator page. This serves the accessibility principle (Section 3): the platform explains what each indicator measures and why it matters.

Profiles. Each county has a profile page (Figure 8) with a real-polygon inset/locator map, a 14-county switcher, metric cards, social sharing, and a standard attribution footer; the locator doubles as a switcher (each county is a link).

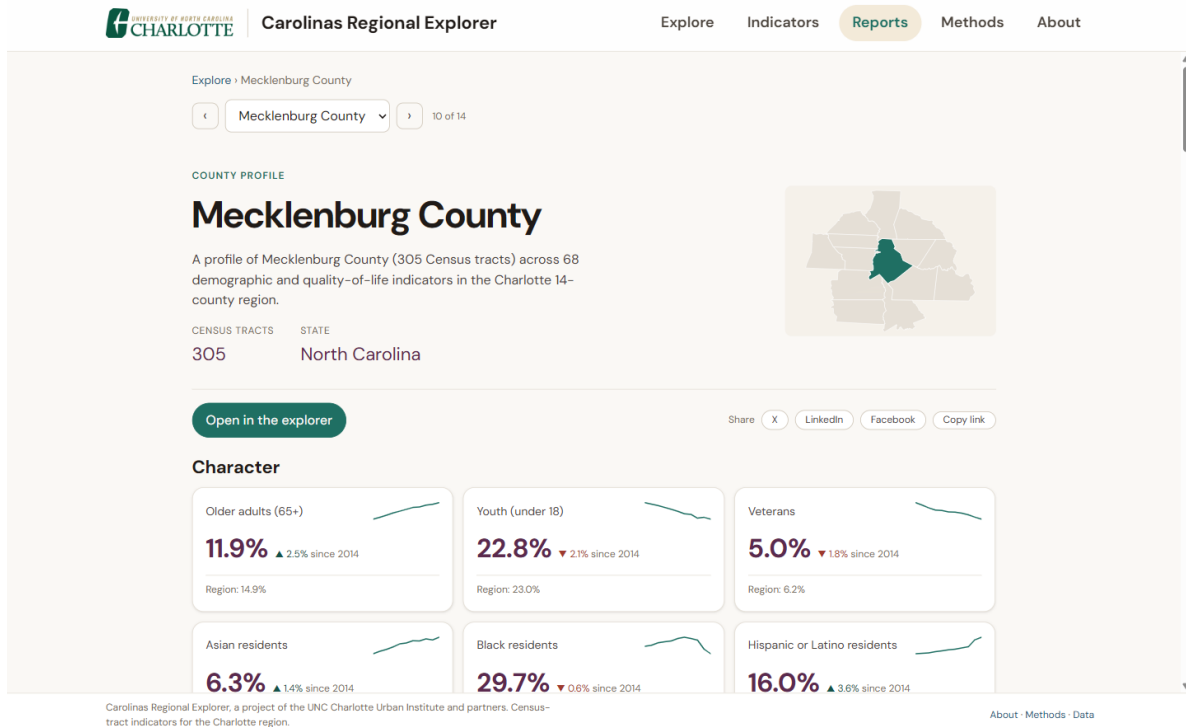


Figure 8: A county profile: a real-polygon locator/inset map (the active county highlighted, every county a link), a 14-county switcher, and metric cards. Profiles and reports carry a standard attribution footer.

Reports and downloads. A custom multi-tract report is printable and shareable on social media (the selection is encoded in the URL), with the attribution footer for any printed or shared artifact. An open-data page (Figure 9) exposes a per-dimension data dictionary with CSV and JSON downloads per indicator and a one-click “Download all” archive; these files are rebuilt with every deploy, so they match the live data.



Carolinas Regional Explorer

[Explore](#)
[Indicators](#)
[Reports](#)
[Methods](#)
[About](#)

OPEN DATA

Data & downloads

Every indicator is open data, published at the 2020 Census-tract level for the 14-county Charlotte region (2001–2024). Download individual indicators or the complete set.

Download data of the explorer

Expand a dimension to see its data dictionary and download each indicator as CSV or JSON.

Character

15

Indicator	Source	Vintage	Download
Asian residents	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Black residents	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Foreign-born residents	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Hispanic or Latino residents	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Households with children	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Households with seniors	U.S. Census Bureau, American Community Survey 5-Year Estimates	2014–2024	CSV JSON
Limited-English-speaking households	U.S. Census Bureau, American Community Survey 5-Year	2016–2024	CSV JSON

Carolinas Regional Explorer, a project of the UNC Charlotte Urban Institute and partners. Census-tract indicators for the Charlotte region.

About · Methods · Data

Figure 9: The data-downloads page: collapsible dimensions reveal a per-indicator data dictionary with CSV and JSON downloads; a single archive bundles the complete dataset.

Throughout, provenance badges (source vintage; *model-based* for CDC PLACES; *pre-2020 harmonized* for ACS) keep the lineage of each number visible (Figure 2).

7. Reproducibility and adoption

7.1. Replication recipe

CRXP is designed so that another region can stand up its own explorer primarily by editing configuration. The high-level recipe:

- Define the region.** Edit `config/region.yml`: list the counties (FIPS), set the tract vintage and CRS, and the ACS years. Comparable-period endpoints are derived automatically from the latest year.
- Install the engine.** Create a Python environment and install the pipeline’s pinned dependencies.
- Provide source access where needed.** Most sources are public and fetched automatically (ACS via the Census API or Census Reporter; CDC PLACES via Socrata; LODES from the LEHD file server; NLCD via the MRLC web service). A few are account-gated or large and are dropped in locally (VIIRS night lights; NLCD Tree Canopy; the census block files for harmonization). Add a free Census API key for multi-vintage history.

4. **Select indicators.** Reuse the provided indicator definitions or add new ones as configuration (choosing a kind: count / median / proportion / rate / density). Bulk additions can be scripted. A local or third-party tract file enters through the generic `tract_csv` source (Section A).
5. **Build.** Run the pipeline per indicator; the QA gate and the printed summary (coverage, comparable-period change and significance, reliability mix) surface problems early.
6. **Hand off and deploy.** Re-run with handoff to merge the data contract into the application, validate it, build the static site, and deploy to any static host. Address search runs as a small serverless function (Netlify in our deployment). Pressing Enter resolves an address to its tract through the U.S. Census Bureau geocoder, which needs no key. Type-ahead suggestions use Geoapify: create a free account at geoapify.com, copy its API key, and set it as `GEOAPIFY_API_KEY` in the Netlify site's environment variables (or in a local `.env` file for `netlify dev`). Without the key, suggestions are off and Enter-to-search still works. Suggestions are limited to the region's bounding box, which the build regenerates from the county geometry.
7. **Document and attribute.** Update the living documentation; retain attribution per the licenses below.

See Section 7.2 for everything a new region edits.

7.2. Region adoption checklist

A new region starts from one file, `config/region.yml`. The excerpt below is the Charlotte file with three of its fourteen counties shown.

```
region:
  name: Charlotte 14-county region
  tract_vintage: 2020          # publish grain: 2020 Census tracts
  crs: EPSG:4326

counties:
  - { fips: '37119', name: Mecklenburg, state: NC }
  - { fips: '37025', name: Cabarrus, state: NC }
  - { fips: '45091', name: York, state: SC }
  # ...

acs:
  product: acs5
  provider: auto               # Census API if CENSUS_API_KEY is set, else Census Reporter
  annual_years: [2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]
  tract2010_years: [2014, 2015, 2016, 2017, 2018, 2019]

reliability:
  cv_caution: 15.0
  cv_unreliable: 30.0
  z_90: 1.645
```

region The region's display name, the tract vintage published (2020), and the coordinate system.
counties Every county by five-digit FIPS code. All tract filtering, county profiles, and regional pooling follow this list.

acs The ACS product and the endpoint years to fetch. Years listed in `tract2010_years` use 2010 tract boundaries and are harmonized to 2020 tracts automatically; the comparable non-overlapping periods are derived from the latest year.

reliability The coefficient-of-variation thresholds for the caution and unreliable flags, and the z-value of the published 90% margins of error.

Table 2 lists what else a region touches. We found these items by searching both code bases for assumptions specific to Charlotte; the items marked *automatic* need no action.

Table 2: Configuration and content a new region edits. Scripts are under `scripts/`; the briefs file is under `src/lib/content/`.

What	Where	Automatic or manual
Counties, ACS years, reliability thresholds	<code>config/region.yml</code>	Manual (one file)
State codes for LODES downloads	<code>config/sources/lodes.yml</code> (states)	Manual
Indicator selection and new tract files	<code>config/indicators/*.yml</code> (Section A)	Manual (one file each)
Census API key (optional, for volume)	<code>CENSUS_API_KEY</code> environment variable	Manual
Tract and county geometry	<code>scripts/gen-fixture.mjs</code> <code>--src <dir></code> , then <code>scripts/gen-geo.mjs</code>	Manual: these scripts were written for Charlotte’s source files and must be adapted
City and town boundaries	<code>scripts/gen-geo.mjs --shp <dir></code>	Manual: supply municipal boundary files; the NC and SC file names and county lists are specific to this region
Neighborhood names	<code>gen-neighborhoods.mjs</code>	Automatic (OSM), or <code>--points</code> with a local file
Map extent and initial view	Fitted to the tract geometry	Automatic
Address-search area	<code>netlify/region.mjs</code> , generated at build	Automatic
Address-search key	<code>GEOAPIFY_API_KEY</code> in the host’s environment	Manual (optional)
Download bundle README (geography, sources)	Generated from the manifest	Automatic
Site text, branding, and logo	Home, About, Methods pages; header and footer	Manual
Regional context panel	<code>RegionContext.svelte</code>	Manual (rewrite or remove)
Plain-language indicator briefs	<code>indicatorBriefs.js</code>	Manual: local context and resources are Charlotte-specific
Default county for Reports	Site navigation (<code>/county/<fips></code>)	Manual
Contact in the OpenStreetMap request header	<code>gen-neighborhoods.mjs</code>	Manual

Running the pipeline needs command-line Python and Node, comfort editing YAML, and a static web host. No desktop GIS software is required.

7.3. Licensing and attribution

The code is released under the MIT License and this paper under Creative Commons Attribution 4.0 (CC BY 4.0); data retain their source licenses (ACS and LODES public domain; CDC PLACES public domain; USGS/NLCD public; OpenStreetMap ODbL; EOG/VIIRS terms). The pipeline’s reproducibility package is published as an open repository ([Venkitasubramanian 2026b](#)), and the application source is public as well ([Venkitasubramanian 2026a](#)). A `CITATION.cff` file gives the canonical citation; an archived Zenodo release will add a DOI. Regional adaptations should credit the original work, as the license requires. In the application, shareable and printable artifacts (profiles, reports) carry a standard attribution footer.

7.4. Living documentation

Three companion documents, an application technical reference, a data-pipeline reference, and a methodology guide, are maintained alongside the code and are supplementary material to this paper.

8. Discussion

8.1. Strengths

CRXP is, to our knowledge, unusual in combining an open application *and* an open pipeline with user-facing uncertainty. The combination matters more than any one part. Because a single versioned contract feeds the map, the profiles, the reports, and the downloads, a number carries the same margin of error, reliability flag, and vintage wherever a user meets it, and a correction to the pipeline reaches every page at the next build.

8.2. Limitations

The platform has so far been validated in a single region, and broader adoption will test its generality. Most of the engine transfers without change; what each region must supply is local (county list, any parcel or local layers, and access to account-gated sources).

Two statistical approximations remain. County metrics are the published county estimates and regional rates are exactly pooled, but the regional value of a *median or index* is a population-weighted approximation (an exact regional median would require microdata). Medians under tract harmonization are likewise approximated as a weighted average of tract medians; a binned-distribution reconstruction is planned.

The data have limits too. Neighborhood attribution is approximate and, with the default OpenStreetMap points, depends on uneven OSM coverage; a region can substitute its own points file.

Several sources are static or account-gated, and raster measures carry classification accuracy rather than sampling error. We list these to invite correction.

8.3. Equity and ethics

Small-area data carries specific risks: disclosure and the suppression of small counts, over-interpretation of uncertain estimates, deficit framing of neighborhoods, and the question of who governs and benefits from the platform. Reliability flags, documented methods, and plain-language briefs reduce some of these risks. Governance is a separate question that software cannot settle; it needs community participation in deciding what is measured and how it is framed.

9. Future work

Planned data extensions include commuting flows and job characteristics from LODES, housing affordability (CHAS), life expectancy, air quality, and food access, and indicators for the empty Safety and Arts & Culture themes (Engagement so far has one residential-stability measure). Methodological work includes small-area estimation from public-use microdata for cross-tabulations the ACS does not publish at tract level (Venkitasubramanian 2026c), the binned-distribution reconstruction of harmonized medians, and cluster detection that accounts for each estimate’s margin of error. On the product side, we plan a composite equity index with propagated uncertainty, demographic-disaggregation views, proximity indicators from parcels and network distance, and a shared regional data backend serving several applications from one warehouse. A container image (Docker) bundling the pipeline and the application build would let adopters run CRXP without setting up Python and Node environments.

10. Conclusion

Communities will keep making decisions on tract-level numbers. CRXP’s contribution is to publish those numbers with their uncertainty and their method attached, in a form another region can run. We publish the platform, its code, and this paper so that other regions can adopt it and reviewers can check it.

11. Reproducibility statement

The data pipeline, its committed warehouse, and its tests are openly available as a reproducibility package (Venkitasubramanian 2026b), released under the MIT License; the application source is public (Venkitasubramanian 2026a), and the live application is at <https://carolinasexplorer.com/map>. An archived release will carry a DOI. The full reproducibility mechanisms (pinned dependencies, committed warehouse, provenance record, and contract validation) are detailed in Section 5.13. This paper is versioned; cite the version and date.

12. References

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A. Worked example: adding a data source end to end

This appendix follows one new dataset from download to map, using FEMA’s National Risk Index (NRI) as the example. The indicator was built in a scratch copy of the platform to produce the listings and screenshots below; it is an illustration and is not part of the live explorer. Adding an indicator from a source the platform already supports is shorter still: it takes one configuration file like `config/indicators/median-age.yml` and the build command in step 3.

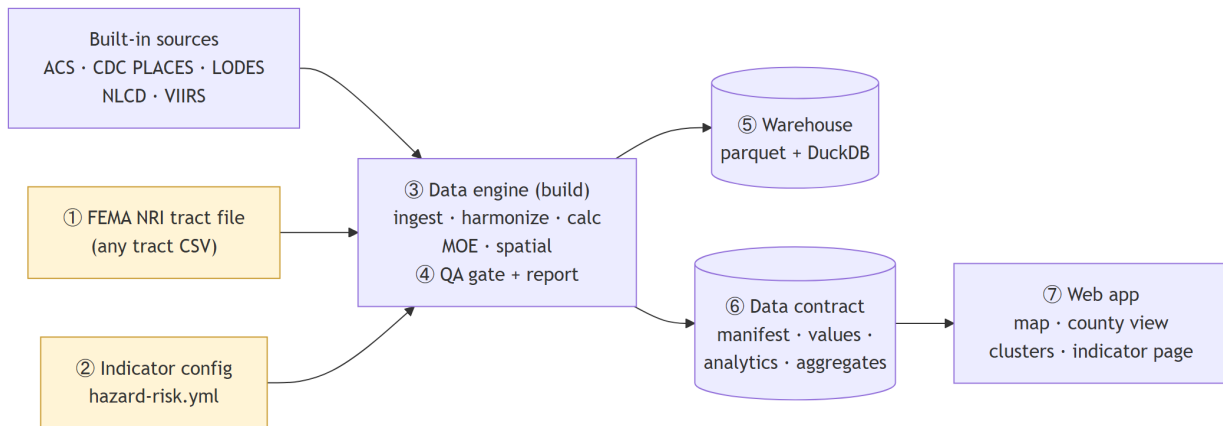


Figure 10: The path a new tract file takes through the platform. The adopter supplies the two highlighted inputs, the file (1) and its configuration (2); the platform does the rest (3–7).

Step 1: get the data. The NRI publishes a composite risk score for every census tract, built from expected annual losses across eighteen natural hazards together with social vulnerability and community resilience. FEMA serves it through a public ArcGIS feature service; the script below downloads the North and South Carolina tracts (v1.20, December 2025) into `data/local/`. Any table with an 11-digit tract GEOID and a value column would work the same way.

```

import requests, pandas as pd

# FEMA National Risk Index (v1.20, December 2025), census-tract layer, NC + SC
URL = ("https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/"
      "National_Risk_Index_Census_Tracts/FeatureServer/0/query")
rows, offset = [], 0
while True:
    page = requests.get(URL, params={
        "where": "STATEFIPS IN ('37','45')",
        "outFields": "TRACTFIPS,COUNTY,STATEABBRV,RISK_SCORE,RISK_RATNG,NRI_VER",
        "orderByFields": "TRACTFIPS", "resultOffset": offset, "resultRecordCount": 2000,
        "returnGeometry": "false", "f": "json"}).json()
    rows += [{"attributes": f["attributes"]} for f in page["features"]]
    offset += len(page["features"])
    if not page.get("exceededTransferLimit"):
        break
pd.DataFrame(rows).to_csv("data/local/nri_tracts.csv", index=False)

# data/local/nri_tracts.csv (first rows)
# TRACTFIPS,COUNTY,STATEABBRV,RISK_SCORE,RISK_RATNG,NRI_VER
# 37001020100,Alamance,NC,40.59077449966109,Relatively Low,December 2025
# 37001020200,Alamance,NC,22.53695313521934,Relatively Low,December 2025
# 37001020301,Alamance,NC,30.224870084311416,Relatively Low,December 2025

```

Step 2: describe it. One YAML file tells the platform what the file contains: which column holds the GEOID and which the value, the data year and tract vintage, the kind of measure, and how to credit the source. The score has no margin of error, so `moe_col` is omitted; a file that has one names it here and gets the same reliability flags as the ACS. Check the file's tract vintage before setting `tract_vintage`: NRI v1.20 uses 2020 tracts, and the build stops with a hint if the GEOIDs do not match the declared vintage.

```

id: 901
slug: hazard-risk
label: Natural hazard risk score
theme: environment
format: number
decimals: 1
higher_is_better: false
source: tract_csv
tract_csv:
  path: data/local/nri_tracts.csv
  geoid_col: TRACTFIPS
  value_col: RISK_SCORE
  year: 2025
  tract_vintage: 2020
  kind: index
  source_label: FEMA National Risk Index v1.20
  source_url: https://www.fema.gov/flood-maps/products-tools/national-risk-index
description: FEMA's composite natural-hazard risk score (0-100; higher means more risk)
meta_why: The score combines expected annual losses from 18 natural hazards with social
          vulnerability and community resilience, so it shows where a disaster would do the most harm.

```

Step 3: build. A single command ingests the file, keeps the region's tracts, runs the QA gate, writes the warehouse, computes z-scores and clusters, derives county and regional values, and merges the result into the application's data contract.

```

$ python -m crxp_data.build hazard-risk --handoff
[1/6] Ingesting tract CSV data/local/nri_tracts.csv (FEMA National Risk Index v1.20) ...
[2/6] Calc (value + optional MOE -> CV/reliability) ...
      kept 752/752 geoids in the 2020 tract set
      QA gate OK (1 years checked; report -> warehouse/qa/hazard-risk.json)
[2/4] Warehouse (parquet + duckdb) ...
      wrote 752 rows -> warehouse/tract_indicators/901.parquet
[3/4] Comparable-period change (non-overlapping pairs) ...
[4/4] App contract (values + analytics + aggregates) ...
      county+region rollout -> 15 rows (warehouse/county_indicators/901.parquet)
      wrote exports/crxp/data/.../901.json (years [2025])
Handoff ...
      merged into crxp/static/data (values, analytics, meta, manifest, aggregates, provenance)
Done.

```

Step 4: check the QA report. The gate's record for the new indicator: all 752 tracts present, no missing values, status **pass**. The same file is regenerated on every build, and a failing check writes it before the build stops.

```

// warehouse/qa/hazard-risk.json
{
  "id": 901,
  "slug": "hazard-risk",
  "label": "Natural hazard risk score",
  "source": "tract_csv",
  "status": "pass",
  "checked_on": "2026-10-05",
  "range": null,
  "years": {
    "2025": {
      "coverage": 1.0,
      "null_rate": 0.0,
      "n": 752
    }
  },
  "problems": []
}

```

Step 5: query the warehouse. The new rows sit in the same long-format table as every other indicator, so any SQL client can read them. **moe** and **reliability** are empty because the source publishes no margin of error.

```

SELECT geoid, year, value, moe, reliability, est_method, source_id
FROM read_parquet('warehouse/tract_indicators/901.parquet')
ORDER BY value DESC
LIMIT 5;

-- result
--      geoid  year  value  moe  reliability  est_method  source_id
-- 45057011209 2025 96.1091 <NA>      <NA>    tract_csv  tract_csv
-- 37179020401 2025 95.1197 <NA>      <NA>    tract_csv  tract_csv
-- 45091061008 2025 90.5545 <NA>      <NA>    tract_csv  tract_csv
-- 37179021020 2025 90.5165 <NA>      <NA>    tract_csv  tract_csv
-- 37119003020 2025 90.0039 <NA>      <NA>    tract_csv  tract_csv

```

Step 6: inspect the contract. The handoff added a manifest entry, which credits FEMA and

records that county values are derived from tracts, and a values file the application reads directly.

```
// manifest.json: entry for this indicator
{
  "id": 901,
  "slug": "hazard-risk",
  "label": "Natural hazard risk score",
  "description": "FEMA's composite natural-hazard risk score (0-100; higher means more risk)",
  "category": "environment",
  "format": "number",
  "decimals": 1,
  "higherIsBetter": false,
  "classMethod": "quantile",
  "years": [2025],
  "geoLevels": ["tract", "county"],
  "source": "FEMA National Risk Index v1.20",
  "vintage": "2025-2025",
  "metaPath": "/data/meta/m901.md",
  "related": [],
  "hasZ": true,
  "hasLisa": true,
  "countyMethod": "derived from tracts"
}

// values/901.json (excerpt)
{
  "indicatorId": 901,
  "years": [2025],
  "values": {
    "37007920100": [33.2537],
    "37007920200": [57.9382],
    "37007920301": [86.4923]
  },
  "...": "749 more tracts, plus breaks and summary statistics per year"
}
```

Step 7: view it in the application. No application code changed. The indicator appears in the Environment theme with its own choropleth, indicator page, county view, and cluster map (Figure 11, Figure 12, Figure 13, Figure 14).

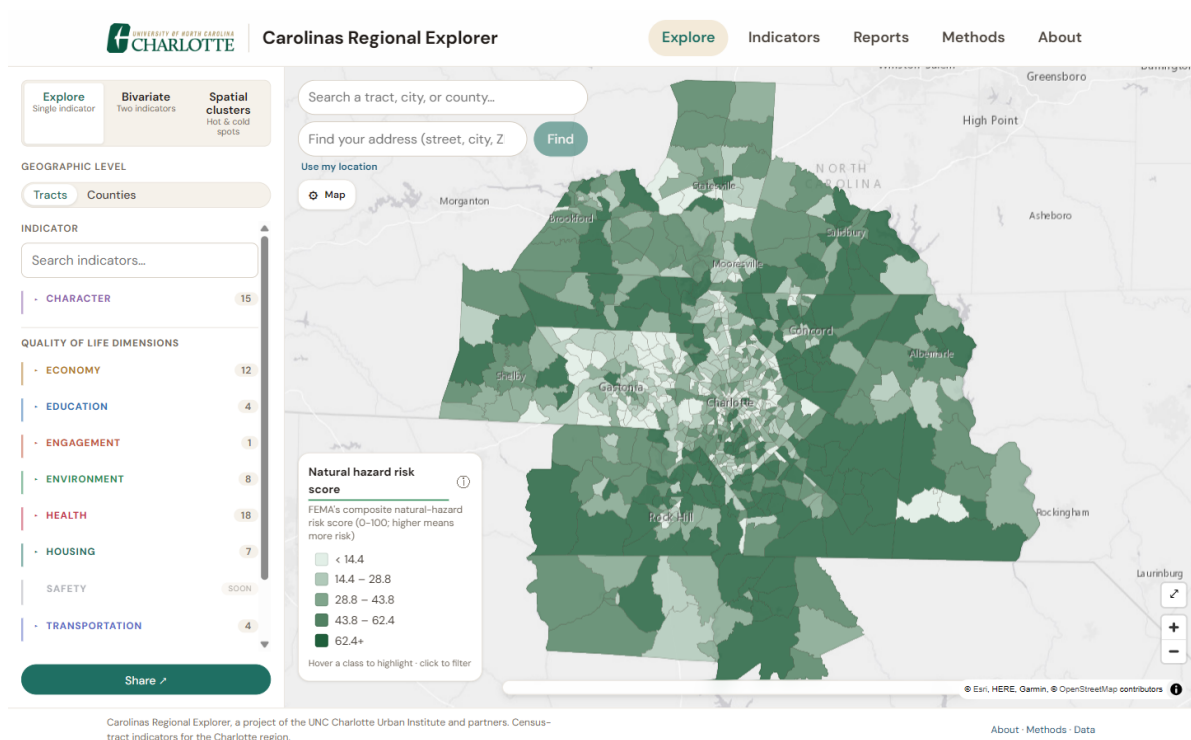


Figure 11: The new indicator on the tract choropleth, classified into quintiles like every other indicator.

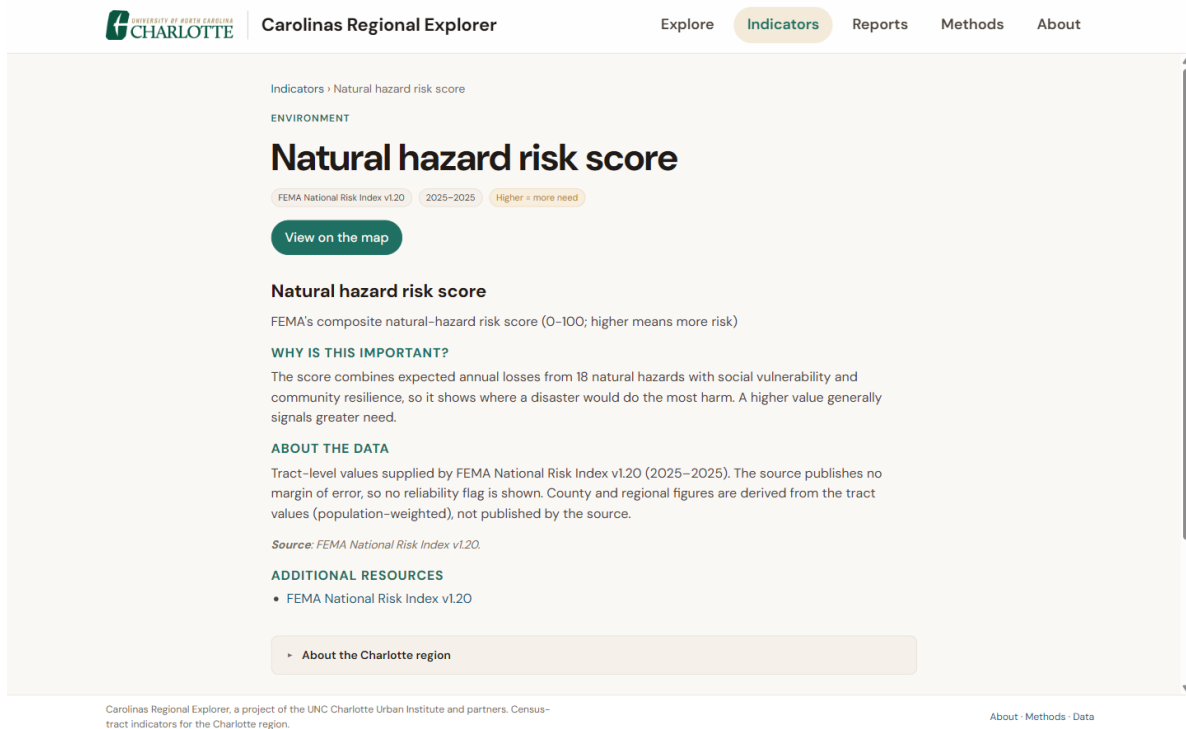


Figure 12: The indicator page generated from the configuration: the FEMA credit, the note that the source publishes no margin of error, and the note that county and regional figures are derived from tracts.

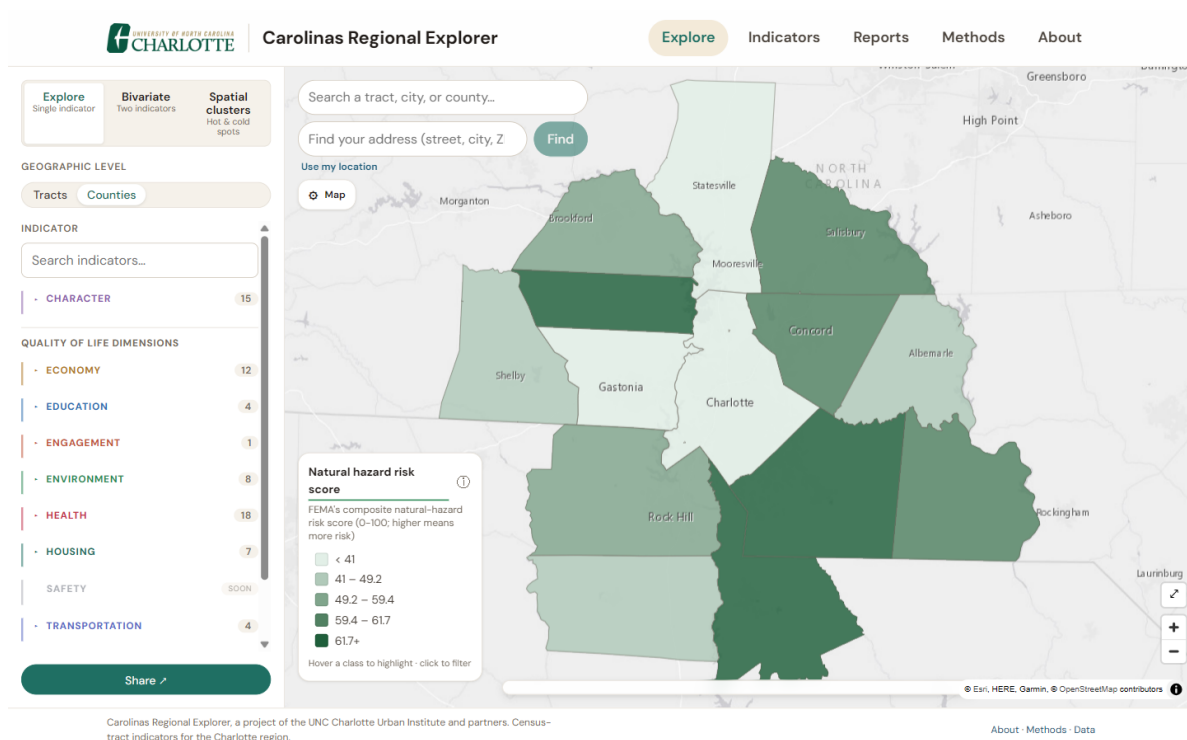


Figure 13: The county view, from county values derived as population-weighted means of the tract scores.

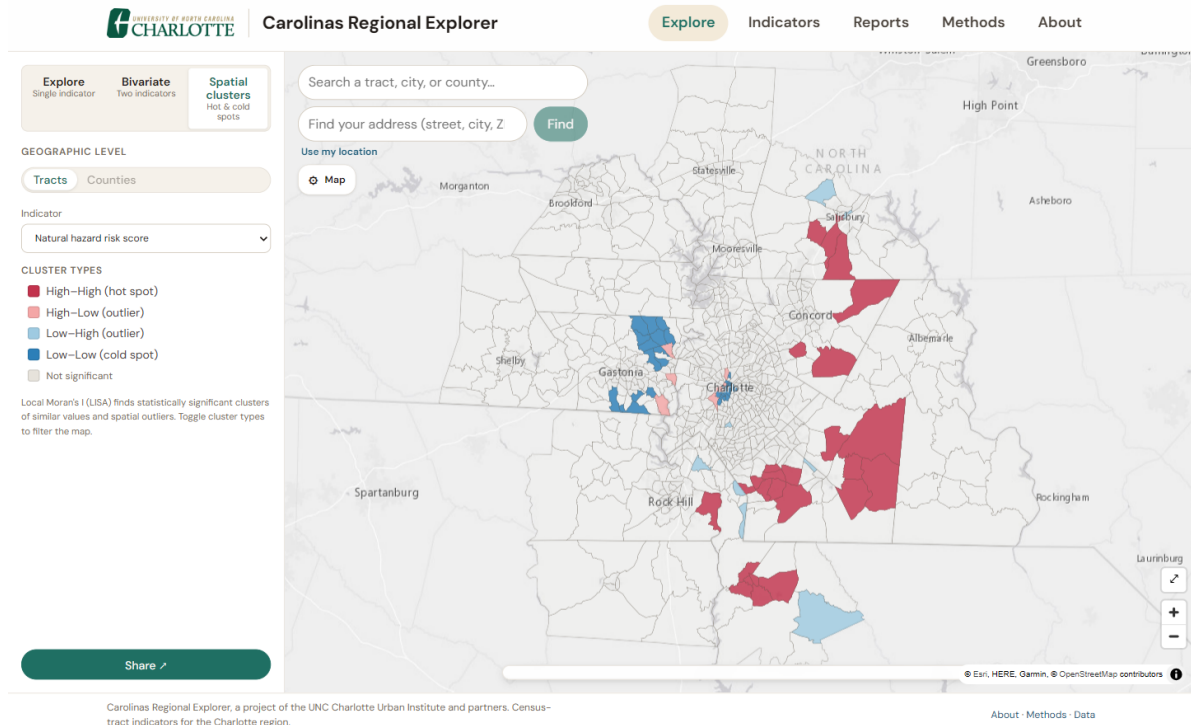


Figure 14: Spatial clusters (LISA) for the new indicator, computed at tract level during the build; the county option is disabled in this mode.

In all, the adopter supplied one CSV file and one YAML file. Quality checks, storage, spatial statistics, county and regional values, metadata, and the map came from the platform.